ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-13 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

Input	-1	0	2
Output	1	3	7

Describe an operation rule that works for every pair.

2. I can use input-output rules to represent relationships.

A machine adds 4 and then doubles the result. The output is 18. Find the input.

3. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

4. I can use input-output rules to represent relationships.

Rules A and B are $2x + 1$ and $x + 3$. At which input do they give the same output?

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Function or not?

$(0, 1), (2, 3), (0, 4)$

6. I can distinguish functions from relationships that are not functions.

Does $\{(1, 2), (1, 3), (2, 4)\}$ represent a function? Explain.

7. I can distinguish functions from relationships that are not functions.

The pairs are $(1, 4), (2, 6), (1, 7)$. Is this relation a function? Explain.

8. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

9. I can connect linear growth in tables, graphs, and equations.

Write an equation.

Slope 4; y-intercept −2.

10. I can connect linear growth in tables, graphs, and equations.

Find the slope.

Points (1,3) and (5,11)

11. I can connect linear growth in tables, graphs, and equations.

A table has $x = 0, 1, 2, 3$ and $y = 3, 7, 11, 15$. Write a rule.

12. I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

13. I can calculate and interpret slope as a constant rate of change.

A tank has 12 L at minute 1 and 27 L at minute 6. Find and interpret its average rate of change.

14. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0, 2)$ and $(3, 8)$.

15. I can calculate and interpret slope as a constant rate of change.

Find the slope through (0,3) and (4,11). Then evaluate $f(2)$ for $f(x) = 3x - 1$.

16. I can calculate and interpret slope as a constant rate of change.

Find the run.

$(-1,8)$ to $(6,10)$

I CAN COMPARE VERTICAL AND HORIZONTAL CHANGES TO FIND SLOPE.

17. I can compare vertical and horizontal changes to find slope.

Find the slope between $(-2, 5)$ and $(2, -3)$.

18. I can compare vertical and horizontal changes to find slope.

Find the slope.

Rise −6, run 3

19. I can compare vertical and horizontal changes to find slope.

Find rise/run from (−2,1) to (2,−3). Then find y when $x = 4$ in $y = 2x + 1$.

20. I can compare vertical and horizontal changes to find slope.

Calculate the slope.

$(0,-5)$ to $(4,3)$